

nano text editor with command:

```
sudo nano /etc/php5/conf.d/wiringpi.  
ini
```

2. Write the following lines in it:

```
extension=wiringpi.so  
wiringpi.pinmaptype=PINS
```

Save it using 'ctrl+o' and exit using 'ctrl+x.'

3. Restart apache server using:

```
/etc/init.d/apache2 restart
```

4. Run the following command to check whether the wiringpi module is loaded or not:

```
php -m | grep wiringpi
```

Software

With WiringPi and PHP already installed, you just have to download the source code (pwm.php) and copy it in '/var/www' directory. Once copied, you can directly run it using the command:

```
sudo php -f pwm.php
```

Now put an LED between pins 6 (ground pin) and 12 to see that its intensity is changing continuously due to a variety of PWM waves. This PWM signal is used in many ways in robotics—one use is to control the speed of motors in the H-bridge by putting this signal at the 'enable' pin.

Project 3: Motor speed control

We started our experiment with GPIO. php but soon we hit its limitations. Then we discovered that Gordon Henderson of the UK made WiringPi software in line with Arduino's Wiring software, which is more versatile than GPIO.php.

WiringPi has the following func-

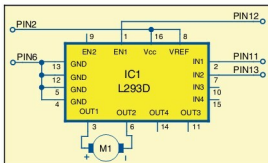


Fig. 4: Circuit for project 3

TABLE II Input Choices and Corresponding Motor Movements

Enter choice	Motion
1	M1 forward start
11	M1 forward stop
2	M1 reverse start
21	M1 reverse stop

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

tions inbuilt:

pwmWrite, pullUpDownControl, digitalWrite, digitalRead, shiftOut, shiftIn, delay, delayMicroseconds, serialOpen, serialClose, serialPuts, serialPuchar, serialDataAvail, serialGetchar, serialPrintf, wiringPiSetupGpio.

With these functions, lots of things can be done before you hit another set of limitations of WiringPi. Till that time, enjoy playing around with Raspberry Pi.

Circuit and working

As shown in Fig. 4, rewrite the circuit that we built around IC1 in project 1. First, remove one set of input/output motor connections. In project 1, enable pins EN1 and EN2 of IC1 were connected to +5 V to make them permanently enabled but since here we will use them as speed control inputs, untie these two wires. Connect pin 1 to physical pin 12 of Raspberry Pi—that's the PWM pin.

Pin 8 of IC1 is for motor power supply. Since we are using a very small coreless motor, we have directly connected it to the +5V supply of Raspberry Pi. In case your motor requires more power, connect it to a separate power supply unit.

Working of the circuit is simple. Once you run the program (explained in software section), you will be asked to enter the direction and speed. Enter your choice as shown in Table II and speed on the scale of 0 to 20.

Software

As you already have WiringPi-PHP installed in project 2, you need not install anything additional now. Download the source code (hbridge-control.php) and copy it in '/var/www' directly. Run the program using the following command:

```
sudo php -f hbridge-control.php
```

Enter the parameters it asks and see the motor change its speed as well as direction. ●

The author is an avid user of open source softwares. Professionally he is a thermal power expert, works as additional general manager at NTPC Limited

Raspberry Pi

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